

MEASUREMENTS AND UNITS

BASE SI UNITS

The SI (Système International d'Unités) is the modern form of the metric system of measurements, and it is used by most countries. It consists of series of units of measurement built around seven interdependant base units of individual physical qualities. All other physical qualities are obtained from these units.

UNIT	SYMBOL	DEFINITION
Meter	m	This unit is the length of a path traveled by light in a vacuum during a time interval of $\frac{1}{299,729,458}$ Of a second.
Kilogram	kg	The unit of mass, equal to the mass of the international prototype kilogram.
Second	s	The second is the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom.
Ampere	A	The constant electric current which, if maintained in two straight parallel conductors of infinite length and negligible cross section, and placed 1 meter apart in a vacuum, would produce between these conductors a force equal to 2×10^{-7} newtons per meter.
Kelvin	K	The unit of thermodynamic temperature, this is the fraction $\frac{1}{273.16}$ of the thermodynamic temperature of the triple point of water.
Candela	cd	The luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of $\frac{1}{683}$ watt per steradian.
Mole	mol	The mole amount of substance that contains as many elementary units as there are carbon atoms in 0.012 kilograms of carbon-12.

SI PREFIXES

SI prefixes and symbols are used to indicate decimal multiples and submultiples of SI units to avoid having to write either very large or extremely small numeric values, from 10^{18} to 10^{-18} . When the number is written, the prefix attaches directly to the name of the unit—for example “nanosecond.” Similarly, a prefix symbol can also be attached to the symbol for the unit—for example “ns.”

FACTOR	PREFIX	SYMBOL	FACTOR	PREFIX	SYMBOL
10^{18}	exa-	E	10^{-1}	deci-	d
10^{15}	peta-	P	10^{-2}	centi-	c
10^{12}	tera-	T	10^{-3}	milli-	m
10^9	giga-	G	10^{-6}	micro-	μ
10^6	mega-	M	10^{-9}	nano-	n
10^3	kilo-	k	10^{-12}	pico-	p
10^2	hecto-	h	10^{-15}	femto-	f
10^1	deca-	da	10^{-18}	atto-	a

SUPPLEMENTARY AND DERIVED SI UNITS

The SI is an evolving system in which units are created and definitions are modified as the technology and precision of measurement improves. In addition to the seven base SI units, there are two supplementary units and many other units derived from the SI base units.

SUPPLEMENTARY UNITS	SYMBOL	DEFINITION
Radian	rad	The unit of measurement of angle; it is the angle subtended at the center of a circle by an arc equal in length to the circle's radius.
Steradian	sr	The unit of measurement of solid angle; it is the solid angle subtended at the center of a circle by a spherical cap equal in area to the square of the circle's radius.

DERIVED UNITS	SYMBOL	DEFINITION
Hertz	Hz	The unit of frequency; 1 hertz has a periodic interval of 1 second.
Newton	N	A unit of force equal to the force that imparts an acceleration of 1 m/sec/sec to a mass of 1 kilogram.
Pascal	Pa	A unit of pressure equal to 1 newton per square meter.
Joule	J	A unit of energy exerted by a force of 1 newton acting to move an object through a distance of 1 meter.
Watt	W	A unit of power equal to 1 joule per second; it is also the power dissipated by a current of 1 ampere flowing across a resistance of 1 ohm.
Coulomb	C	A unit of electrical charge equal to the amount of charge transferred by a current of 1 ampere in 1 second.
Volt	V	A unit of potential equal to the potential difference between two points on a conductor carrying a current of 1 ampere when the power dissipated between the two points is 1 watt; it is equivalent to the potential difference across a resistance of 1 ohm when 1 ampere of current flows through it.
Farad	F	A unit of capacitance of a capacitor that has an equal and opposite charge of 1 coulomb on each plate and a voltage difference of 1 volt between the plates.
Ohm	Ω	A unit of electrical resistance equal to the resistance between two points on a conductor when a potential difference of 1 volt between them produces a current of 1 ampere.
Siemens	S	A unit of electrical conductance, the reciprocal of 1 ohm: 1 divided by 1 ohm.

